Perspective-Aware Billboard Corner Detection with a Web-Based End-to-End System

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Abstract

Billboard placement technologies have become increasingly important in digital advertising, multimedia production, and augmented reality applications. Accurate computation of the four corner coordinates of a billboard are essential for the seamless integration of new billboard content. We formulate a loss function to minimise the error across the computed four corner coordinates while accounting for varying perspective alignments. We evaluate the accuracy of the corner coordinate predictions using the Root Mean Square Error (RMSE), obtaining scores of 4.0682 for x-coordinates and 3.8553 for y-coordinates. These results indicate a high level of accuracy in corner coordinate computation. Next, this paper presents a web-based automated billboard system (ABS) that integrates video upload, billboard detection, localisation, and placement within a unified end-to-end framework. The method's practical applications extend to media companies and digital content creators, offering valuable insights to video editors and broadening its impact.

Keywords: Digital advertising, Corner coordinates detection, Billboard detection, Web application

1 Introduction

Advertisements in multimedia videos play a crucial role in capturing the attention of a broad audience. Recent interest focuses on integrating advertising boards with modern techniques [Krishna and Aizawa, 2018], enhancing effectiveness in the evolving digital advertising landscape. For effective billboard integration, accurate detection and corner coordinate computation are crucial. Traditionally, video editors manually inspect image frames and mark the opposite corners to insert new advertisements, making it a time-consuming and labor-intensive process. This technological gap significantly affects industries such as advertising and sports broadcasting, as most existing research approaches billboard detection and analysis as independent tasks, thereby constraining the development of fully integrated intelligent advertising systems that can jointly detect, analyse, and optimise billboard placement in real-world environments.

The primary contribution of this paper is the development of a novel corner detection algorithm. We have also extended this to design and implement an end-to-end automated billboard system that supports practical user interaction, video processing, and advertisement management workflows. This work focuses on the system-level integration and deployment of existing billboard detection [Dhang et al., 2024a] models within a scalable web-based application framework. This is particularly crucial in applying the billboard system, as end users are often media companies, video editors or content creators. The main contributions to this work [1] are: *i)* Extended traditional corner detection with proposed Corner Detection for Advert (CDANet), enabling precise corner computation under different perspective alignment. *ii)* Defined a custom loss function that minimises

[1] To facilitate the reproducibility of this research, the code of this paper is made available: https://github.com/sukritidhang/CEDA-Net-cornercoord-detection.git

error across all four corners, improving localisation accuracy to support multiple billboards in a video frame. *iii)* Design and implementation of a user-friendly web-based automatic billboard system interface, Provide real-time preview and interactive editing functionalities for billboard analysis and visualisation.

The rest of the paper is organized as follows: Section 2 provides the background and related work. Section 3, outline the proposed methodology. Section 4 presents evaluation results. Finally, Section 5 concludes the paper by summarising the results.

2 State of the Art

Earlier edge detection methods relied on first-order [Sobel et al., 1968] and second-order [Torre and Poggio, 1986] operators, which were limited to predicting axis-aligned bounding boxes (x_1, y_1, x_2, y_2) along with the width (w) and height (h) of the object. Difficulties arise when the perspective alignment of the object deviates from the conventional axis $(x_1, y_1, x_2^{'}, y_2^{'}, x_3, y_3, x_4^{'}, y_4^{'})$. These perspectives aid in comprehensively understanding an object's dimensions and spatial relationships by learning its bounding box, making accurate computation of corner points for effective image registration [Dhang et al., 2025]. Recent deep learning methods have improved object detection and edge prediction [Liu, 2023]. Wibisono *et al.* [Wibisono and Hang, 2020] combined traditional edge detectors with CNNs, while RCNN [Girshick et al., 2014] and Faster RCNN [Ren et al., 2015] introduced regression-based bounding box refinement. RetinaNet [Lin et al., 2017] further enhanced general object detection, but such methods primarily focus on standard objects and rectangular bounding boxes. Dhang *et al.* [Dhang et al., 2024b] proposed a billboard segmentation framework that localises billboards and extracts the corner points. The method employs Otsu's thresholding, followed by erosion and dilation to reduce noise and reconnect fragmented areas, with contour detection used to identify the final billboard boundaries. This model computes billboard corner coordinates rather than estimating perspective-aligned billboard corner coordinates.

Despite advances in complex model architectures, scalability and deployment remain major challenges. Smart advertisement insertion has gained attention, though most studies address detection, classification, segmentation, and inpainting as independent tasks. For instance, Bulkan *et al.* [Bulkan et al., 2023] proposed a smart advertisement insertion framework with duration constraints. Cloud-based systems such as AdNext [Akgul et al., 2020] depend heavily on CNNs and remote infrastructure, limiting suitability for latency-sensitive environments, whereas lightweight systems like LiveSense [Chen et al., 2019] improve scalability using pre-trained multimodal models and reduced cloud dependency. Despite these advances, robust deployment across diverse devices and network conditions remains difficult. Collectively, these works indicate a shift toward intelligent and context-aware advert insertion systems.

3 Methodology

3.1 Dataset

In this work, we utilised the ALOS dataset [Dev et al., 2019], to train the CDA model, we chose 1,750 images as the positive class (images containing advert) and 1,750 images as the negative class (images without advert) [Heonho, 2022][Bansal, 2019]. For evaluation, four publicly available non-copyrighted advert videos were randomly selected from YouTube, covering varying viewpoints and angles. Due to the limited availability of advert-focused video datasets with ground-truth corner annotations, the evaluation was conducted on this small custom sample. The description of the dataset is summarised in Table 1.

3.2 Proposed Corner Detection for Advert (CDA) model

The perspective alignment of the object deviates from the conventional axis $(x_1, y_1, x_2^{'}, y_2^{'}, x_3, y_3, x_4^{'}, y_4^{'})$, introducing complexities in corner coordinate computation. Understanding these types of perspective viewpoints is crucial, as they offer different angles for observing and representing an object. These viewpoints include:

Table 1: Dataset description

Name	ALOS Dataset	Video Dataset			
		Video 1	Video 2	Video 3	Video4
Total number of images	9315	2874	1231	1573	4022
Dimension	512 x 512	1280 x720	1280 x720	1070 x 720	1280 x720
Color grading	RGB	RGB	RGB	RGB	RGB
Billboard images	1750	1333	-	1573	2710
No-billboard images	1750	1514	1231	-	1312

i) Front View *ii)* Side View *iii)* Perspective View. The Figure 1 depicts the different viewpoints of the billboard class.

(a) Front view (b) Side view (c) Perspective view

Figure 1: Represents different types of perspective viewpoint.

We adopt pretrained AdSegNet model [Dhang et al., 2024b] to generate the probabilistic mask. The model employs an encoder-decoder architecture, where the encoder comprises three convolution layers with filters [64, 128, 256] and three max-pooling layers sized (3×3), and the decoder includes four deconvolution layers and three up-sampling layers. The output of the encoder block is fed into the decoder block with deconvolution layers consisting of filters [256, 128, 64]. The final deconvolution layer comprises a filter with dimensions (1×1) for the localization output. To enable billboard corner prediction, the decoder output was flattened and connected to an additional fully connected regression module. This module consists of two FC layers with 1024 neurons and ReLU activation, with a dropout layer (rate = 0.5) applied after the first FC layer. The final FC layer contains eight neurons with sigmoid activation, representing the normalised coordinates of the four billboard corners. These modifications allow the network to directly regress corner points. To accurately predict the coordinates, we formulate the loss function as follows: Let $g = [x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4]$ denote a ground-truth coordinates, represented by left top coordinate (x_1, y_1), right top coordinate (x_2, y_2), left bottom coordinate (x_3, y_3) and right bottom coordinate (x_4, y_4). The predicted coordinates as $\hat{p} = [\hat{x}_1, \hat{y}_1, \hat{x}_2, \hat{y}_2, \hat{x}_3, \hat{y}_3, \hat{x}_4, \hat{y}_4]$. The mean squared error (MSE) for each individual coordinate (x and y) is calculated separately, and then these individual losses are summed to accurately predict each of the four coordinates independently.

The loss function used for computing corner coordinates provides valuable insight, as it is designed to minimize the error across all four corner coordinates. Unlike standard regression losses that focus on individual coordinates (such as the top-left or bottom-right corners), this loss function considers all four corners of the bounding box, which is crucial for accurate and complex perspectives and varying positions. The total loss is the sum of the squared errors between the predicted and ground truth coordinates for each corner. For each corner, the squared difference between the predicted and actual x-coordinates and y-coordinates is computed, and these errors are averaged over the dataset. The formula for corner's loss is defined as $loss_{x_1} = \frac{1}{N}\sum_{i=1}^{N}(\hat{x}_{1_i} - x_{1_i})^2$, $loss_{y_1} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_{1_i} - y_{1_i})^2$. Similarly, we calculated the $loss_{x_2}, loss_{y_2}, loss_{x_3}, loss_{y_3}, loss_{x_4}, loss_{y_4}$. We define the total loss function of the CDA model by the following equation:
loss_coordinates = $(loss_{x_1} + loss_{y_1} + loss_{x_2} + loss_{y_2} + loss_{x_3} + loss_{y_3} + loss_{x_4} + loss_{y_4})$.
By considering all four corners, the model can better account for non-axis-aligned bounding boxes, ensuring accurate predictions even when billboards are in varying visibility levels (minority billboard area, and majority billboard area) and different perspective alignment.

3.3 Automatic Billboard System (ABS) architecture

Next, a automated billboard system has been designed. This section outlines the development of a responsive, efficient system design that leverages React for the front-end and Python for the back-end services. Figure 2 demonstrates the flow diagram of the ABS. This end-to-end system integrates all necessary components, such as data processing, detection, localisation and integration, into one cohesive pipeline. In this web application, the Billboard Video Upload (BVU) component was designed for uploading videos. Uploaded videos are temporarily stored within the web application, allowing users to manage video content directly through the interface. In addition, the system provides video preview capabilities and displays real-time video metadata such as file format, resolution, duration, and file size.

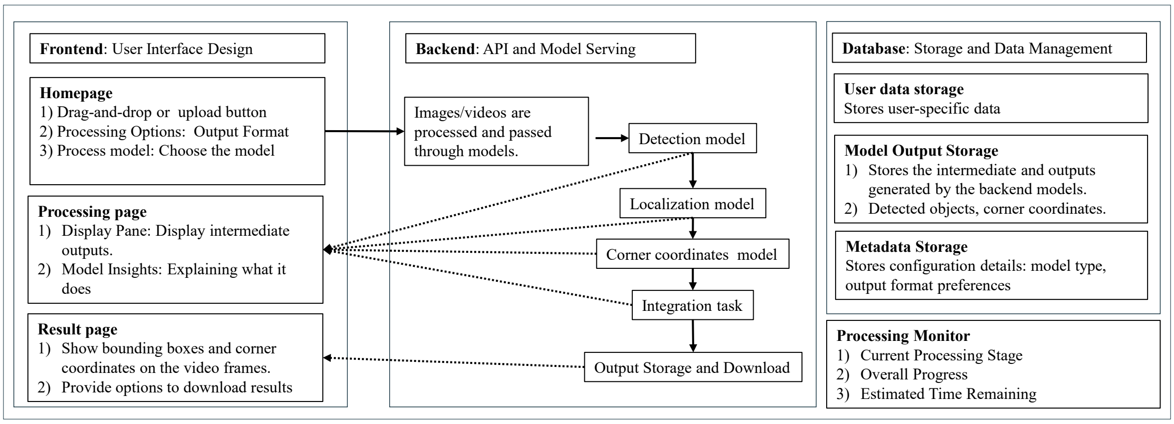


Figure 2: Automated billboard placement system architecture

3.3.1 System Architecture

The system architecture adopts a frontend-backend model to support efficient user interaction, real-time processing, and seamless integration with deep learning-based billboard detection models.

Front-end Technology: The frontend of the application was developed using React and Ant Design component library. React was selected due to its component-based architecture, which is highly suitable for building complex and interactive user interfaces. In addition, React's virtual Document Object Model (DOM) allows efficient rendering and updating of interface components, which is particularly beneficial for real-time previewing and interactive billboard manipulation tasks within the application. The Ant Design framework was integrated to provide a comprehensive collection of high-quality user interface components with consistent styling and responsive layouts. Furthermore, the framework improves the overall user experience through modern design principles and extensive documentation support. Figure 3 illustrates the front-end interface used for BVU and processing.

Back-end Technology: The backend system was implemented using Python for computer vision, and deep learning applications. Furthermore, Python offers seamless implementation with deep learning frameworks enabling efficient deployment of the billboard detection models proposed in [Dhang et al., 2024a, Dhang et al., 2024b]. A RESTful API architecture was adopted to establish standardised communication between the front-end and back-end systems. The API-based design improves modularity, maintainability, and scalability while allowing the front-end interface to communicate efficiently with the detection and processing modules.

4 Experimental results

Our evaluation includes both static images and video frames, demonstrating the model's ability to compute corner-coordinates of the billboard under various visibility levels and different perspective alignments. We then evaluated the performance of the ABS.

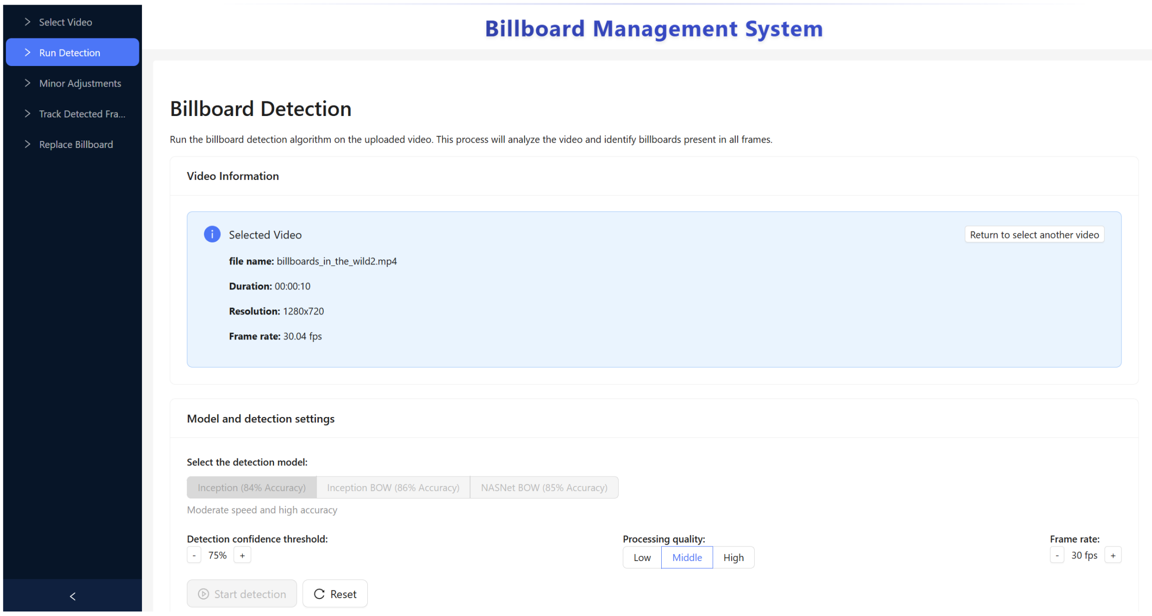


Figure 3: Interface for billboard video upload

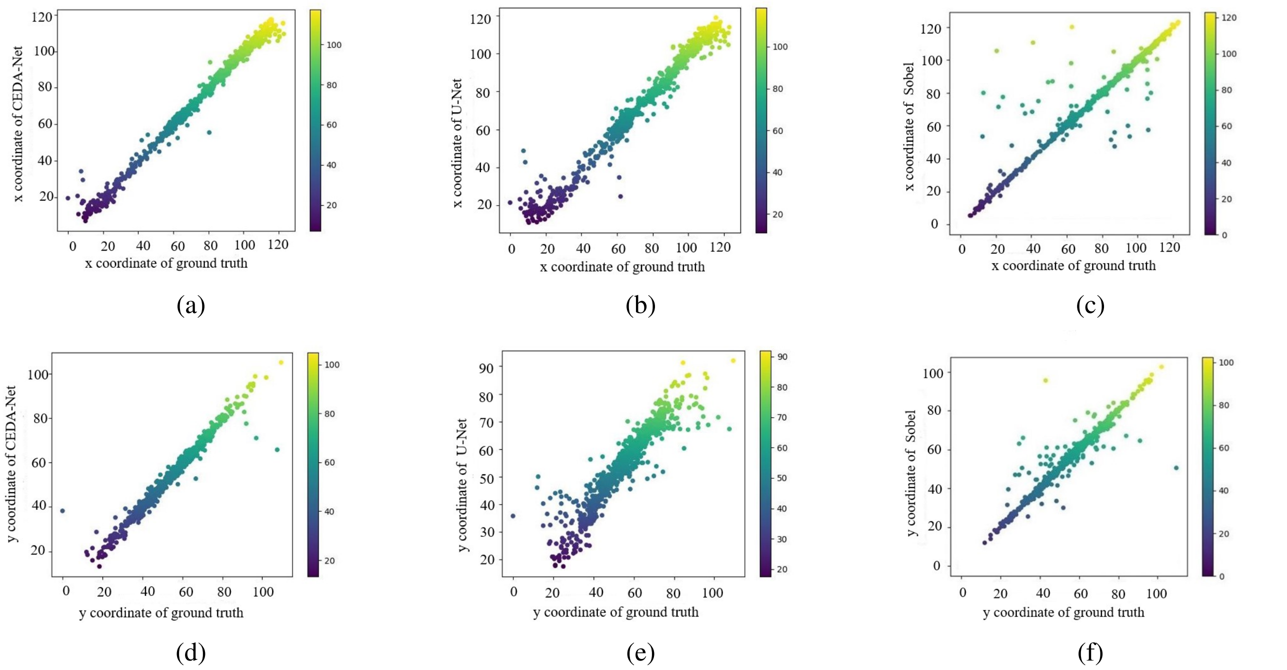
4.1 CDA model performance

We assess the accuracy of the predicted (pred) four corner coordinates against the ground truth (gt) corner coordinates, by calculating the root mean square error (RMSE) metric as expressed RMSE $= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2}$, the square root of the mean of the squared differences between the pred and gt coordinate values. where, x_i is the average of all the four corner coordinates along x-axis of the ground truth and $\hat{x}_i$ is the average of all the four corner coordinates along x-axis of the pred coordinates, n is the number of testing images. Similarly, RMSE is calculated for y-axis. We also determine the correlation coefficient between the pred coordinates and gt coordinates using the equation $r = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(\hat{x}_i - \overline{\hat{x}})}{\sqrt{\sum_{i=1}^{n}(x_i-\overline{x})^2 \sum_{i=1}^{n}(\hat{x}_i - \overline{\hat{x}})^2}}$ where, $\overline{x}$ is the mean of the four corner coordinates of the gt and $\overline{\hat{x}}$ is the mean of the pred four corner coordinates.

Table 2: Summarisation of RMSE and correlation coefficient of models, with custom loss function and without custom loss function on ALOS dataset (All results are average across the dataset).

Model	RMSE ↓		Correlation coefficient ↑	
	X coord	Y coord	X coord	Y coord
With custom loss function				
Proposed CDA	**4.0682**	**3.8553**	**0.9875**	**0.9647**
UNet	6.0244	7.1055	0.9696	0.8553
Without custom loss function [Dhang et al., 2024b]				
AdSegNet	**17.7350**	**15.3519**	**0.8257**	**0.6416**
UNet	22.9443	18.6021	0.6988	0.4845
Sobel edge detection method	17.7350	15.3519	0.8257	0.6416

Table 2 presents the results obtained with and without the proposed custom loss function. Compared with AdSegNet, the proposed CDA model demonstrates superior performance in corner coordinate prediction. The proposed CDA reduces the RMSE by 77.06% and 74.88% for the x- and y-coordinates, respectively. Additionally, the correlation coefficients improve by 19.5% for the x-coordinate and 50.35% for the y-coordinate. These improvements indicate that the proposed CDA model provides more accurate corner coordinates that are closer to the gt coordinates. Figure 4a compares the average gt x-coords (in x-axis) with the pred x-coords from the proposed CDA model (in y-axis). Figure 4b and 4c show the comparison between the average gt x-coords and the pred x-coords using the U-Net and Sobel methods, respectively. Similarly, we plot the average pred y-coords obtained using the proposed CDA model, U-Net and Sobel methods in Figure 4d, 4e and 4f, respec-



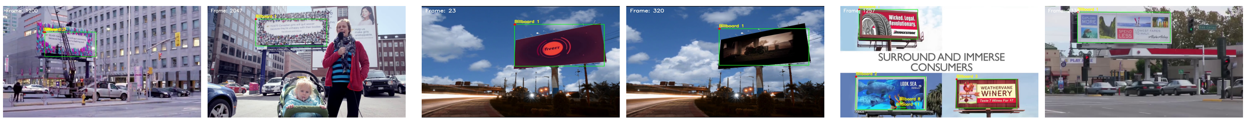
(a) (b) (c)

(d) (e) (f)

Figure 4: Representation of correlation coefficient plot. We represent (a) x-coords of proposed CDA vs ground truth coords, (b) x-coords of U-Net vs ground truth coords, (c) x-coords of sobel detection method vs ground truth coords, (d) y-coords of proposed CDA vs ground truth coords, (e) y-coords of U-Net vs ground truth coords, (f) y-coords of sobel detection method vs ground truth coords.

(a) Video 1 (b) Video 3 (c) Video 4

Figure 5: Frame-level extraction of billboard corner coordinate by UNet model.



(a) Video 1 (b) Video 3 (c) Video 4

Figure 6: Frame-level extraction of billboard corner coordinates using the proposed CDA model.

tively. The selected representative cases in Figure 5 and 6 reflect billboards in different perspective alignment and visibility levels to ensure that evaluation covers a wide range of conditions for videos.

4.2 ABS Performance

This section focuses on system integration, operational functionality, interface usability, and deployment performance within the proposed web-based framework. The system successfully processes uploaded videos by extracting frames at configurable intervals and performing billboard localisation on the extracted frames. Figure 7 depicts the billboard detection interface displaying test results, confidence score, and whether the billboard exists.

The backend API communication operated reliably through testing, enabling efficient data exchange between the front-end and deep learning models. Resource utilisation monitoring tracked CPU, memory, and

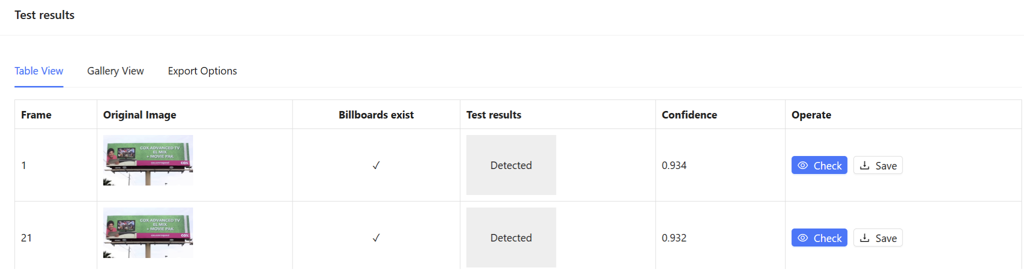


Figure 7: Billboard detection interface with results display

Table 3: ABS: Resource utilisation summary

Operation	CPU	Memory	Network
Video Upload	15–25%	120–180 MB	10–50 MB/s
Billboard analysis	70–90%	1.2–1.8 GB	5–10 MB/s
UI Rendering	5–15%	150–250 MB	0.1–1 MB/s
Idle Application	1–3%	90–120 MB	< 0.1 MB/s

The color intensity indicates the relative resource demand of each operation, where:

Red denotes high utilisation.

Yellow denotes moderate utilisation.

Green denotes low utilisation.

network usage of ABS during operation of the implemented components as tabulated in Table 3, confirming that most operations maintain reasonable resource usage. Thus, ABS represents the most computationally intensive process due to the high processing demands of the deep learning model.

5 Conclusion

Billboard analysis has become increasingly important in digital advertising and multimedia production. We formulate a loss function to minimise the error across the computed four corner coordinates while accounting for varying perspective alignments. Our findings indicate predicting corner coordinates in video frames with RMSE scores of 4.0682 for x-coordinates and 3.8553 for y-coordinates, a high level of accuracy in corner coordinate computation. We also presented a web-based automated system that integrates video upload, frame extraction, billboard detection, interactive editing, and result presentation into a unified pipeline.

Overall, the proposed system demonstrates the feasibility of integrating automated billboard placement and management into a scalable web-based platform, offering practical benefits for digital advertising and multimedia editing. Although large-scale annotation of billboard corner coordinates is labour-intensive, future work will focus on expanding the evaluation using more comprehensive advertisement video datasets with ground-truth annotations.

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